



Stationary model of laser-induced plasma: Critical evaluation and applications

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ABSTRACT

We present an algorithm for both laser-induced plasma emission spectra simulation and rapid peak-identification suitable for selection of analytical lines (free of spectral interferences, minimally self-absorbed, most intense lines). Calculations of plasma composition and radiation transport are based on the model of a uniform plasma under local thermodynamic equilibrium (LTE) requiring $3 + (K - 1)$ parameters – temperature T , electron density n_e , plasma mass m and array of K mass fractions of chemical elements $\{c\}$. A simple cost function based on Pearson's r with a sole maximum in T – n_e space is proposed to achieve the best convergence between the experimental and model spectra. We have found the best fitted T , n_e , m for spatially integrated experimental spectra of standard samples of high- and low-alloy steels at delays of 1–10 μ s (gate 0.4 μ s) after a laser pulse within three spectral ranges (255.1–276.4, 392.6–412.2, 529.3–547.0 nm) by a brute force technique. The instrumental function of the Czerny–Turner spectrometer equipped with an intensified charge coupled device (ICCD) detector has been accurately accounted for which has been notably improved the model and experimental spectra similarity. Several optimizations including a fast Fourier transform (FFT) convolution and a polynomial approximation of Voigt profile allowed acceleration of computing to several milliseconds per spectrum.

We observed a discrepancy between the best-fitted values of T obtained in the ultraviolet (UV) and visible ranges, while the n_e values were close to each other in these ranges. The UV range is dominated by the ionic Fe II lines but most of the lines in the visible range are atomic Fe I lines. Thus, a possible explanation of the T difference is an inhomogeneity of the plasma (different temperatures for the “hot” plasma core and the “cold” periphery). As a spectacular example for line identification, we demonstrate that 275 emission lines are ascribed to 106 peaks within the UV range out of a total of ~12,000 lines found in spectral databases. Selection of analytical lines to determine chromium in cast iron and carbon in steels is discussed. Applicability of the algorithm to reveal analytical lines for silver, lanthanum and yttrium determination in soils and ores is mentioned. The developed algorithms are useful for choice of the best analytical line under certain plasma and spectrometer parameters in terms of: (i) spectral interferences, (ii) transmittance and (iii) intensity, for estimation of detection limits of trace elements in a known sample matrix, for detection of erroneous entries in databases, including transition probabilities and Stark parameters.

1. Introduction

Laser-induced plasma spectra modeling is useful in several applications: (1) investigation of analytical performance/capability of a specific analysis of a given sample using laser-induced breakdown spectroscopy (LIBS); (2) calibration-free analysis of an unknown sample; (3) detection of erroneous entries in databases (transition probabilities or Stark parameters values) or unknown blending lines leading to huge apparent discrepancies between the model and experimental spectra. The first key point involves automation of the peak-

identification procedure (a time-consuming task in the case of a complex sample matrix) in the experimental spectra [1] and, furthermore, evaluation of interferences for selection of the best analytical line.

Although unique analytical capabilities of LIBS are obvious [2], the route from the sample to the analytical signal is rather complex [3]. A comprehensive physical modeling of the laser-induced plasma starting from laser-matter interaction to radiation transport in plasma is complicated by a variety of originating processes. Consequently, the modeling is commonly separated into two tasks: (1) laser ablation (laser-matter interaction, mass removal, ionization/breakdown and laser-

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plasma interaction) [4,5]; (2) laser-induced plasma expansion into the ambient medium (formation of shockwaves, partitioning of energy, mass and radiation transfer) [6]. However, there was an attempt of a simultaneous modeling of these two parts within rough approximations [7]. The ablation part requires the transverse energy distribution in the laser beam and many additional material parameters as inputs. Generally modeling of the ablation is limited to the case of pure metals irradiated by a Gaussian laser beam. The plasma expansion part requires initial conditions in the plume at the moment of the laser pulse termination (distributions of temperature and mass density), which cannot be strictly verified in the experiments.

A hydrodynamic model of the laser-induced plasma based on conservation of mass, momentum and energy equations was presented in 2004 by Igor Gornushkin for the case of spherical symmetric plasma expansion into vacuum [8]. The model was consistently complicated to describe an asymmetric expansion into vacuum [9], symmetric expansion into the ambient gas (shock waves phenomena and percolation of ambient gas into the plasma was included) [10]. Its current version involves transport phenomena (viscosity, thermal conductivity, diffusion, sample surface adhesion) and considers the plasma as real gas during its axisymmetric expansion into the ambient gas [6]. The final version was also supported by accurate calculations of radiation transport taking into account the plasma and collection optics geometries [11]. Unfortunately, implementation of the model requires extensive spatially resolved measurements of the plasma followed by Abel or Radon transformation of the experimental data. Such measurements [12–14] are confined by the small sizes and short lifetime of the plasma plume. However, in LIBS practice we commonly have spatially integrated spectra of the plasma plume. It is mostly the closeness of the experimental and model spectra under certain experimental conditions that is highly desirable for analytical applications. Thus the practice arises the following questions: (i) Is it possible to better explain the observable spectra including the additional physics and chemistry phenomena into the model? (the so-called forward task), and (ii) Can we accurately and uniquely deduce the plasma parameters from the experimental spectral data? (the so-called inverse problem). Gornushkin et al. [15] found the initial temperature and element number densities in a plasma within the approximation of its spherical expansion into vacuum (the radius of the plasma was determined from a plasma image) from spatially- and time-integrated spectra of an aluminum alloy using a simulated annealing optimization algorithm. A good agreement between experimental and model spectra (Pearson's correlation coefficient $r = 0.96\text{--}0.99$) was obtained within the narrow spectral band 279–288.5 nm, while the optimization procedure was not performed within the full broad band 220–300 nm due to the high time cost of the calculations. In a subsequent study, Gornushkin et al. [10] found only a qualitative agreement between the model and experimental spectra of SiC within the range 280–290 nm for the plasma expanding into Ar at an atmospheric pressure after an extensive variation of initial plasma parameters (number densities of the constituents, temperature, plasma radius, and speed of the plasma edge). Calibration-free analysis can serve as a test of advantages of the dynamic modeling in comparison with the stationary plasma model. Fitting of experimental spectra by synthetic ones calculated in approximation of plasma expansion into vacuum resulted in similar uncertainties (30%–250%) for the predicted elements mass fractions in aluminum alloys [16] in comparison with calculations made with the well-known approach of Cuicci et al. [17] for the stationary plasma under local thermodynamic equilibrium (LTE). Results of quantitative comparisons of the most complete model [6] of plasma expansion with experimental data, to our best knowledge, have not yet been published in peer-reviewed literature. Obviously, both the high-cost computations for plasma dynamics modeling and the time-consuming and complicated procedure of laser-induced plasma tomography needing specific equipment lead to insufficient experimental confirmation of the model. Due to these reasons the hydrodynamic part of the model as well as

non-uniformity of plasma parameters distribution were omitted for calibration-free analysis of steel slags simulators by fitting of experimental spectra with the help of massive Monte-Carlo spectra calculations on GPU [18].

A one-dimensional dynamic model of laser ablation in gas based on Euler equations has been developed by Colonna et al. [19]. The model provides time-dependent profiles of mass density, temperature, and Mach number. The kinetic processes including radiative recombination of Ti II and Ti III, their three-body recombination with electrons, and Ti ionization were also incorporated into the model [20]. The measured temporal evolution of Ti I and Ti II molar fractions was in quite good agreement with the calculated values. Furthermore, the model was extended to 2D case for both the presence and absence of chemical equilibrium [21].

Due to complexity of the dynamic plasma model, modeling of a static laser-induced plasma (its snapshot at a certain moment after the laser pulse) under LTE seems more appropriate for practical applications. A plasma can be considered as the stack of several uniform zones with different values of the parameters (temperature, electron density, plasma size(s) or volume, or mass and atomic or mass fractions of the elements) to simulate plasma non-uniformity. Hermann et al. [22] obtained more accurate values of temperature (in comparison with the Boltzmann plot method) and electron density of a plasma induced on a Ti target in a low-pressure N_2 atmosphere by comparing the computed line profiles of three Ti II lines with the experimental ones. Dividing the plasma volume into two zones (“hot” core and “cold” periphery) they clearly demonstrated that the self-absorption of spectral lines played an important role in the early dense plasma phases, even for transitions between highly excited energy levels. After including diatomic molecule formation due to chemical reactions with the ambient gases (O_2 , N_2) [23], the plasma model was implemented to investigate the degree of the plasma uniformity in argon and in air from spatial integrated measurements of Ba and Na spectral lines shapes [24]. Furthermore a calibration-free analysis of glasses and an Al alloy was performed by experimental spectra fitting [25,26]. It was shown [26] that the element fractions in the plasma remain the same in time when the electron density is still high enough to provide the necessary condition for LTE existence according to the McWhirter criterion [3]. However, it should be mentioned that, surprisingly, the principal details of spectra fitting (a cost function, initial parameters, constraints, optimization algorithm, etc.) were scarcely described or even omitted. In addition, the obtained composition was not compared with precisely measured values provided by ICP-OES/MS, AAS, etc.

Wester and Noll [27] described a plasma by a stationary spherical shell model (plasma consisting of two zones) surrounded by an ambient gas, which partially absorbs the emitted radiation. A good agreement between self-reversed experimental and calculated Al I profiles (394.401, 396.152 nm) was achieved, while only a satisfactory agreement was obtained for the iron spectrum within the range 531–541 nm. Wester and Noll attributed the discrepancies between simulation and experiment to the low accuracy of the available Stark parameters within the considered spectral range. Finally, those authors demonstrated the suitability of the model for prediction of absorption of the plasma emission by the surrounding gas shell (different concentrations, different path length) in the vacuum UV range.

Returning to the raised question on quality of explanation of the experimental spectra and deducing the plasma parameters we can outline the following issues. Broadening of spectral line profiles by plasma and the instrumental function degrade resolution of the lines in the experimental spectra [28]. This problem is compounded by low accuracy of transition probabilities and mostly by absence of Stark parameters. These can be partially addressed by considering numerous lines in a wide spectral range, which in turn requires: (i) a fast massive modeling with numerous lines, (ii) a stress on the instrumental function and (iii) a robust line identification.

The main goal of our study is a development of a tool for relatively

simple emission spectra modeling and line identification that can be useful for LIBS applications as well as in other fields of analytical atomic spectrometry. The tool should provide a good agreement with experiments within the assumptions made. A freely available Web interface for Atomic Spectra Database (ASD) of the National Institute of Standards and Technology allows simple simulations of atomic spectra ([29], first released in June 2017). However, the algorithm used in this simplified modeling ignores the main disturbing factors, namely, self-absorption of emission lines and Stark broadening and shifts, considering the spectral resolution power R (FWHM of a Gaussian instrumental function) only. Although the database completion is ongoing, it should be noted that a number of transition probabilities, especially for transition metals are absent in the NIST database.

2. Theoretical

2.1. Common assumptions for plasma model

Our initial assumptions for the laser-induced plasma spectra modeling are the following:

- (1) plasma is under LTE conditions [30], uniform and stationary;
- (2) plasma is electrically neutral;
- (3) element mass fractions in the plasma are equal to the ones in the solid sample;
- (4) complete matter atomization (i.e., without consideration of chemical reactions, molecule and cluster formation).

Therefore, we can use $3 + (K - 1)$ parameters for description of the plasma to calculate the densities of the species and the characteristic plasma size (optical path length): (i) temperature ($T = T_{exc} = T_e = T_H \neq T_\nu$, where T_{exc} is excitation temperature, T_e and T_H are kinetic temperatures of electrons and atomic species, and T_ν is photon temperature), (ii) electron density (n_e), (iii) plasma mass (m), (iv) array of K mass fractions of the elements $\{c\} = \{c_1, c_2, \dots, c_K\}$ ($K - 1$ independent mass fractions).

2.2. Calculation of plasma species densities

To calculate densities of the plasma species under LTE, the following parameters should be given: T , n_e and $\{c\}$ (see Section 2.1). Hereafter, we denote the total atomic species number density in the plasma as n (m^{-3}), the total number density of the k -th element species as n_k (m^{-3}) and the number density of the k -th element species with the charge z as $n_{k,z}$ (m^{-3}). We can firstly calculate the ionic species fractions of each element $\beta_{k,z} = n_{k,z}/n_k$ from the system of Saha equations for consecutive ionization states, $z = 1, 2, \dots, Z_k$. For the T in eV we have:

$$\frac{n_{k,z} n_e}{n_{k,z-1}} = 2 \left(\frac{\sqrt{2\pi m_e q_e T}}{h} \right)^3 \frac{U_{k,z}(T)}{U_{k,z-1}(T)} e^{-\frac{E_{k,ion}^{(z-1)}}{T}} \quad (1)$$

where m_e and q_e are mass (kg) and charge (C) of the electron, h is the Plank constant (J·s), $E_{k,ion}^{(z-1)}$ is the ionization potential (eV) of the species with the charge $z - 1$, U is the partition function:

$$U_{k,z}(T) = \sum_i g_{k,z,i} e^{-\frac{E_{k,z,i}}{T}} \quad (2)$$

where $E_{k,z,i}$ (eV) and $g_{k,z,i}$ are energy and the degeneracy of the i -th level retrieved from the NIST ASD [31]. The solution of consecutive Saha equations for the k -th element is similar to the solution used in the NIST LIBS database [26] and also can be found in the supplementary material (File S15, sections 1 and 2). We have neglected the formation of the negative ions in the laser-induced plasma due to low electron binding energy for anions (for example, 0.75 eV for H^- [30]). The ionization lowering factor was also omitted in the Saha Eq. (1) because of its ambiguous calculation [30] and dependence on plasma temperature

and degree of ionization (the last point requires numerical iterative solution of both the Saha and charge neutrality equations in the common case of multi-element, multi-ionized plasma [8]). There are several other points for neglecting of ΔE . For example, different cut-off criteria for summation in the partition function lead to various estimations for the reduction factor. According to Lochte-Holtgreven book [30], ΔE can vary from 0.09 eV for Debye-Hückel theory to 0.85 eV for Brunner's model at $T = 1.6 \times 10^4$ K and $n_e = 2 \times 10^{17} \text{ cm}^{-3}$.

For a given element k , its mass fraction n_k is related to n as:

$$n_k = n \frac{\frac{c_k}{M_k}}{\sum_{k=1}^K \frac{c_k}{M_k}} \quad (3)$$

where M is the atomic weight (g/mol). Using the plasma charge neutrality condition

$$\sum_{k=1}^K \sum_{z=1}^{Z_k} z \beta_{k,z} n_k = n_e \quad (4)$$

and Eq. (3) we can write the total plasma species density as:

$$n = \frac{n_e \sum_{k=1}^K \frac{c_k}{M_k}}{\sum_{k=1}^K \left(\frac{c_k}{M_k} \sum_{z=1}^{Z_k} z \beta_{k,z} \right)} \quad (5)$$

Thus, the number density of a particular species $n_{k,z,i}$ (m^{-3}) at the i -th energy state can be given by the Boltzmann distribution as:

$$n_{k,z,i}(T, n_e, \{c\}) = n_e \frac{c_k}{M_k} \beta_{k,z} \frac{1}{F_{sum}} \frac{g_{k,z,i} e^{-\frac{E_{k,z,i}}{T}}}{U_{k,z}(T)} \quad (6)$$

where the parameter F_{sum} (mol/g) is:

$$F_{sum}(T, n_e, \{c\}) = \sum_{k=1}^K \left(\frac{c_k}{M_k} \sum_{z=1}^{Z_k} z \beta_{k,z} \right) \quad (7)$$

2.3. Plasma emission coefficient

The emission coefficient in the laser-induced plasma under LTE is formed by free-free transitions (the so called bremsstrahlung radiation caused by acceleration of free electrons in the electric field of ions), free-bound transitions (caused by recombination of free electrons with ions) [32, ch. 5], and by bound-bound electronic transitions in atomic species (molecules were not taken into consideration in our model, see Section 2.1). The first two appear as continuous and the last one as discrete spectral radiation. We consider bound-bound transitions only, because the model is intended primarily for analytical LIBS applications. Commonly, analytical measurements are performed at relatively long delays ($\geq 0.5 \mu\text{s}$) after the laser pulse, when the electron density and temperature are within the intervals 10^{16} – $10^{17.7} \text{ cm}^{-3}$ and 0.4–1.5 eV, respectively. Noll has demonstrated [33] that the bremsstrahlung emission coefficient is approximately four or more orders of magnitude lower than the one for the Fe I 538.34 nm line in an iron plasma at delays of 0–8 μs after the second laser pulse in a double-pulse LIBS configuration. The recombination emission is only one order of magnitude greater than the bremsstrahlung emission at $T \sim 0.7$ eV. Therefore, the plasma emission coefficient ϵ_λ ($\text{W}/(\text{sr m}^3 \cdot \text{m})$) can be approximated by the sum of the all of the lines emission coefficients:

$$\epsilon_\lambda(T, n_e, \{c\}) = \sum_{k,z,i} \frac{h \nu_{ij}^{(k,z)}}{4\pi} A_{ij}^{(k,z)} n_{k,z,i} P_{\lambda,ij}^{(k,z)} = \frac{q_e}{4\pi} \sum_{k,z,i} \Delta E_{ij}^{(k,z)} A_{ij}^{(k,z)} n_{k,z,i} P_{\lambda,ij}^{(k,z)} \quad (8)$$

where $\Delta E_{ij}^{(k,z)}$ is the photon energy (eV) corresponding to the transition between upper (i) and lower (j) energy levels, $A_{ij}^{(k,z)}$ is the transition probability (s^{-1}) and $P_{\lambda,ij}^{(k,z)}$ is the line profile (m^{-1} , see Section 2.4 below).

2.4. Line profile

The line profile P_λ is assumed to be normalized [30]:

$$\int_0^\infty P_\lambda d\lambda = 1 \quad (9)$$

We took into consideration the main mechanisms of lines broadening in laser-induced plasma to calculate P_{λ} : Stark and Doppler broadening [33], and we neglected the other ones (natural, pressure, Van der Waals, resonance) due to their small contribution to the resulting profile [34]. Thus, we equated the Lorentzian component (full width at half maximum, FWHM) of line profile to Stark broadening $w_L = w_{\text{Stark,FWHM}}$ (nm) and the Gaussian one to Doppler broadening $w_G = w_{\text{Doppler}}$ (nm). We have neglected the ion contribution to the Stark broadening and shift and used a simplified equation for the quadratic Stark effect to calculate the width w_{Stark} (nm) and shift d_{Stark} (nm) of non-hydrogenic species lines:

$$w_{Stark,FWHM} \approx n_e w_e, \quad d_{Stark} \approx n_e d_e \quad (10)$$

where w_e is the electron impact parameter (nm/cm^{-3}) and d_e is the electron Stark shift parameter (nm/cm^{-3}). Theoretical parameters from the Stark-B database [49] can be fitted to a particular temperature using empirical equations suggested by Sahal-Br  chet et al. [35], while the experimental values of w_e and d_e are usually obtained only for a given temperature (see, for example, reviews by Konjevi   et al. [36,37]). For hydrogenic atoms, the linear Stark effect occurs, and w_{Stark} becomes proportional to $n_e^{2/3}$ according to Griem [50]. We have retrieved the corresponding coefficients of proportionality by approximation of the results of ab initio calculations of w_{Stark} for different n_e by Gigoso and Carde  no [38] at the temperature $T = 10,000 \text{ K}$ (mid value for laser-induced plasma). For example, we have obtained at the reduced mass $\mu = 1.0$ for the Balmer H α line (656.3 nm):

$$w_{\text{Stark},FWHM,H\alpha} = 4.68 \times 10^{-12} n_e^{2/3} \quad (11)$$

FWHM for the Doppler broadened line is given by the following equation [33] (for T in eV):

$$w_{\text{Doppler}} = \left(\frac{8q_e T \ln 2}{Mc^2} \right)^{0.5} \lambda_{L,c} \quad (12)$$

where M is the mass of a species (kg), c is a speed of light in a medium (m/s), and $\lambda_{L,c}$ (nm) is the central line wavelength taking into account a possible Stark shift:

$$\lambda_{L,c} = \lambda_{L,c,Database} + d_{Stark} \quad (13)$$

When the above-mentioned broadening mechanisms act together, the resulted profile becomes a convolution of two functions named the Voigt profile function (in m^{-1}) [39]:

$$P_\lambda = 10^9 \frac{2 \ln 2 w_L}{\pi^{1.5} w_G^2} \int_{-\infty}^{+\infty} \frac{e^{-t^2}}{\ln 2 \left(\frac{w_L}{w_G} \right)^2 + \left(\sqrt{4 \ln 2} \frac{\lambda - \lambda_{L,C}}{w_G} - t \right)^2} dt \quad (14)$$

Since we should calculate the profile of each line, the task can be very time-consuming for a wide spectral region containing thousands or more lines. To boost a performance of our algorithm, we have used several simplifications:

- 1) The weak lines giving relative contribution smaller than the desired level of accuracy δ (commonly, 10^{-5}) are excluded from the further calculations, i.e. the integral emission coefficient of the line $\varepsilon_{k,i}$ ($\text{W}/(\text{sr}\cdot\text{m}^3)$) should meet the criterion:

$$\varepsilon_{k,z,i}/\max_{k,z,i}\varepsilon_{k,z,i}\geq\delta, \quad \varepsilon_{k,z,i}=\frac{q_e}{4\pi}\Delta E_{ij}^{(k,z)}A_{ij}^{(k,z)}n_{k,z,i} \quad (15)$$

- 2) The empirical equation suggested by McLean et al. [40] is used to calculate the Voigt profile instead of a direct numerical integration of Eq. (14). The maximal relative error $\sim 10^{-4}$ in the profile

approximation is acceptable for our application. Nevertheless, the spectra modeling program can directly integrate Eq. (14) by means of the trapezoid rule at the user request within the numerical accuracy of a personal computer (PC) ($\sim 10^{-15}$).

- 3) The region for calculation of the profile, which is starts from the central line wavelength, is also limited by the accuracy level: $P_{\lambda}/P_{\lambda_c} \geq \delta$. P_{λ} is assumed to be equal to zero at the other wavelengths.
- 4) The wavelength step $\Delta\lambda$ (nm) for the profiles (and spectrum) calculations is chosen as a compromise between the accuracy and performance of calculations. Since the step and, partially, resolution of our experimental spectrum was determined by the intensified charge coupled device (ICCD) detector pixel width (see Eq. (25) below), we defined $\Delta\lambda$ in terms of the number of calculated points per pixel. It was empirically established that 10 calculated points per pixel was enough, which corresponds to $\Delta\lambda \approx 0.00142$ nm at $\lambda = 400$ nm for our setup.
- 5) Due to the symmetry of the Voigt profile, we calculate only one half of it.
- 6) We do not calculate the line profiles for all of the lines. Instead of this the “default” line profile specific for each chemical element is implemented. That is, for the lines with an unknown Stark width (most of the lines) the same default user-defined parameter is used. Also, the same Doppler width calculated at the spectral region central wavelength λ_c at a given temperature for line(s) of a specific element with a specific mass was used within a narrow spectral region (for example, changes in $w_{Doppler}$ will be up to 5% within the range 390–410 nm). Of course, if the wavelength within the range of interest greatly varies, we should recalculate the profile because of strong changes in the Doppler line width. And, surely, we calculate the specific profile for the lines with the known Stark width.

2.5. Plasma size and equation of radiative transfer

We used the simple approximation of the plasma plume shape by a 1D uniform “rod” with the length l (m) and cross section l^2 (m²), i.e. the optical path length (characteristic plasma size) can be estimated from the plasma volume V (m³) as:

$$l = \sqrt[3]{V} \quad (16)$$

Multiplying both parts of the charge neutrality Eq. (4) by V and taking into account that the total number of the k -th element species in the plasma $N_k = n_k V = N_A m_k c_k / M_k$ (m is the total mass (g) of the plasma see seq. 2.1), where N_A is the Avogadro constant (mol^{-1}), we further can obtain for V :

$$V = \frac{1}{n_e} \sum_{k=1}^K \sum_{z=1}^{Z_k} z \beta_{k,z} \frac{c_k m}{M_k} N_A = \frac{m N_A}{n_e} F_{sum} \quad (17)$$

It should be noted that either l or V can served as the model parameter instead of m .

The one-dimensional equation of radiative transfer [30] along the line of sight is:

$$\frac{dI_\lambda(x)}{dx} = \varepsilon_\lambda(x) - \kappa(\lambda, x)I_\lambda(x) \quad (18)$$

where $I_\lambda(x)$ is the emission intensity ($\text{W}/(\text{sr m}^2\text{m})$) of the plasma “rod” at the point x (m) along the coaxial line of sight, $\varepsilon_\lambda(x)$ is the emission coefficient ($\text{W}/(\text{sr m}^3\text{m})$) and $\kappa(\lambda, x)$ is the absorption coefficient (m^{-1}), both at the point x . In the case of a uniform plasma with the constant values of ε_λ and $\kappa(\lambda)$ within the plasma “rod”, the equation can be analytically solved. Integration of Eq. (18) over x from 0 to l (optical path length, see Eqs. (16) and (17)) with the boundary condition $I_\lambda(0) = 0$ gives:

$$I_\lambda(l) = \frac{\varepsilon_\lambda}{\kappa(\lambda)}(1 - e^{-\kappa(\lambda)l}) \quad (19)$$

ε_λ can be calculated for a given plasma parameters (T , n_e , $\{c\}$, see

Section 2.3), while $\kappa(\lambda)$ can be found from Kirchhoff's law of thermal radiation:

$$\frac{\varepsilon_\lambda}{\kappa(\lambda)} = B_\lambda(T) = \frac{2hc_0^2}{n_r^2 \lambda^5} \frac{1}{e^{\frac{hc_0}{n_r \lambda q_e T}} - 1} \quad (20)$$

where B_λ is the black-body radiation intensity (W/(sr·m²·m)) given by the Planck law [30] (for T in eV), c_0 is the speed of light in vacuum (m/s), n_r is the wavelength-dependent refractive index of the medium (we can assume $n_{air} \approx 1$ here), λ is the light wavelength in the medium (m). This law is valid under full thermodynamic equilibrium (FTE) as well as under LTE [30,41]. Since emission of laser plasma is commonly observed in an air environment, the tabulated air wavelengths [31,47] under standard conditions (dry air at 15 °C, 101325 Pa pressure, and with 450 ppm CO₂ content) are used for spectra calibration and calculations, and the air refractive index equation for the wavelength region 185–1700 nm was retrieved from five-parameters formula (3) in the Ref. [42]. Thus, we have a simple equation for the plasma emission intensity:

$$I_\lambda(l) = B_\lambda(1 - e^{-(\varepsilon_\lambda/B_\lambda)l}) \quad (21)$$

2.6. The instrumental function

The instrumental profile strongly affects the plasma spectrum to be experimentally observed. Since we directly compare an experimental spectrum with the model one in our applications (see Section 4), the convolution of the model spectrum $I_\lambda(l)$ (Eq. (21)) with the instrumental function $P_{\lambda',Instr}$ (nm⁻¹) is required:

$$S_\lambda = \int P_{\lambda',Instr} I_{\lambda-\lambda'} d\lambda', \quad \int P_{\lambda',Instr} d\lambda' = 1 \quad (22)$$

$P_{\lambda',Instr}$ is a result of a convolution of the spectrograph function $P_{\lambda',Spectrograph}$ and the detector one $P_{\lambda',ICCD}$:

$$P_{\lambda',Instr} = \int P_{\lambda',Spectrograph} P_{\lambda'-a,ICCD} da \quad (23)$$

We have calculated the reciprocal linear dispersion (RLD) and $P_{a,Spectrograph}$ for the Czerny–Turner spectrograph commonly used in LIBS (as well as in our system, see Section 3) as a convolution of the slit and diffraction functions (see Supplementary Material, File S15, eqs. S15.8 and S15.9); the spectrograph aberrations were unknown and not taken into account. The detector function for the ICCD camera is unknown in a general form due to the camera design and the image intensifier features (photocathode and microchannel plate materials, channel diameter, back fiber-optic plate resolution, phosphor material, etc.). Generally, the manufacturer provides informative value in units of line pairs per mm (lp/mm) to characterize image intensifier spatial resolution in accordance with 1951 USAF Resolution Test Targets (or its analogues). Retrieval of the $P_{\lambda',ICCD}$ for our camera will be discussed further below in Section 3.2.

After the convolution of a spectrum with the instrumental function, we averaged the spectral intensity within the spectral pixel width of the CCD $w_{\lambda,pix}$ (nm) and discretized the intensity values according to the bit width of the analog-to-digital converter (ADC) (see Supplementary Material, File S15, eq. S15.16).

3. Experimental

3.1. Experimental setup and samples

A common LIBS laboratory experimental setup was used in the present study. The radiation of the second harmonic (532 nm) of the Q-switched Nd:YAG laser (LS 2134UTF, LOTIS TII, Belarus, $\tau \sim 6$ ns, Q-switch was operated at 5 Hz) was directed by mirrors and prisms and focused by an achromatic doublet lens (NUDL 30150P, SigmaKoki, Japan, $f = 151.5$ mm) at 6 mm below the sample surface for better

Table 1
Spectrograph and camera parameters.

Parameter	Value
Slit width, s	25 μ
Collimator f.d., f_{coll}	320 mm
Objective f.d., f_{obj}	320 mm
Spectrometer angle, α	24°
Groove density, N_d	1800 mm ⁻¹
Grating width, B	48 mm
Diffraction order, m_d	1
Intensifier resolution, w_{ICCD}	32 lp/mm
Focon ratio, FR^a	1.52
CCD pixel width, w_{pix}	6.45 μ
CCD width, N_{CCD}	1390 pix.
ADC bit width, bit	12 bit

^a FR is a scaling coefficient (dimensionless) corresponding to an image transferred from the back surface of the intensifier to the CCD plane by the fiber optic taper.

reproducibility. The spot diameter was ~ 0.3 mm, the pulse energy was controlled by a pyroelectric sensor (ES220C Thorlabs, USA) and was equal to 52 ± 1 mJ, which resulted in the power density ~ 12 GW/cm². The central part of the plasma plume was projected onto the 25 μ slit of the 32 cm Czerny–Turner spectrograph (HR-320, USA, $R \sim 7000$ at 400 nm) by a two-lens condenser (the axis of the laser beam and the plasma plume were parallel to the slit). The spectra were recorded by an ICCD camera Nanogate-2 V (Nanoscan, Russia; details can be found elsewhere [43]) and were axially integrated. We have summarized the spectrograph and the camera parameters in Table 1, since they were used for spectra modeling.

The relative spectral sensitivity of the setup was measured with the use of the deuterium and tungsten-halogen calibrated lamps (SL-3 CAL, SL-1 CAL, Stellarnet, USA) within the spectral range 205–920 nm (Fig. 1). The two deep local minima on the curve can be attributed to the materials of the intensifier photocathode. The registration system had a very poor sensitivity outside the range 220–900 nm and had no response below 205 nm and above 940 nm.

The analyzed set of the samples included low-alloy steel (UG112, ZAO ISO, Russia) and high-alloy steel (CRM 475, BCS, UK) (Table 2). We performed the measurements of the steel samples in four spectral windows: 255.094–276.352, 392.535–412.221, 529.29–546.96, 648.08–663.54 nm (the uncertainty of the wavelength calibration was smaller than 0.005 nm for the first two ranges, and smaller than 0.01 nm for the last two ones). Spectra were recorded at delays of 1, 2, 3, 4, 5, 7, and 10 μ s after the laser pulse for the first three ranges and at 1, 1.5, 2, 2.5, 3, 4, 5, and 7 μ s for the fourth one. The gate was fixed at a relatively short value (250 ns) to neglect variations of plasma

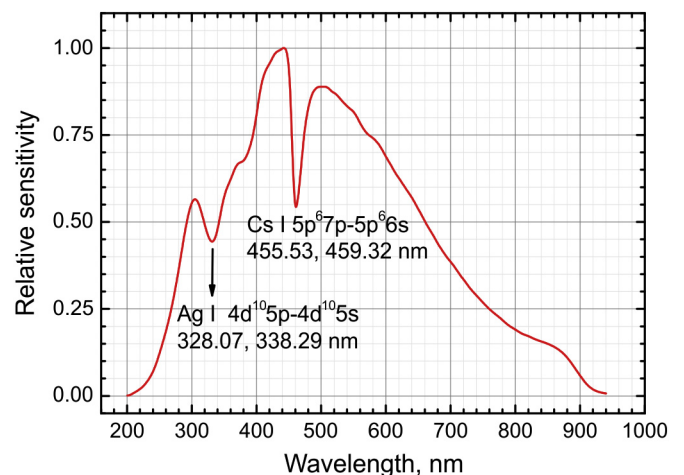


Fig. 1. Relative spectral sensitivity of the measurement system.

Table 2
Standard samples composition.

Element	c, mass. %	
	UG112	CRM 475
C	0.186	0.050
Al	0.026	0.013
Si	0.60	0.21
P	0.0065	0.037
S	0.005	0.008
Ti	0.0028	–
V	0.014	–
Cr	0.98	14.14
Mn	1.63	0.89
Co	–	0.22
Ni	0.185	5.66
Cu	0.157	1.94
Zr	0.0047	–
Nb	–	0.22
Mo	0.021	1.59
Sn	–	0.015
W	0.005	–
Fe	~96.2	~75.0

parameters within the registration time window. Each spectrum was averaged over $n_{meas} = 10$ consecutive measurements at the same spot, while each measurement itself was directly accumulated on the detector from 5 consecutive laser shots to enhance the signal-to-noise ratio. The average relative standard deviation of the mean pixel intensity ($RSD_i/n_{meas}^{0.5}$) for the spectrum was ~1.7–2.0%, while maximum values of RSD_i (at the “backgrounds points”) did not exceed 4%. The averaged spectra were corrected for the relative spectral sensitivity of the system (Fig. 1), and a constant background equal to the minimum intensity was removed from the spectra. The spectral region around 656 nm was used for the plasma electron density estimation only with the use of the H_α Balmer line, while the correlation of the model and experimental spectra (see Section 3.3) was performed within the rest three ranges.

3.2. ICCD instrumental function measurements

Since the instrumental function strongly affects the observed line shape and has no complete theoretical description (due to unknown spectrograph aberrations and the image intensifier function, see Section 2.6), we have determined it experimentally with the help of a low pressure Ar–Hg lamp (SL-2, Stellarnet, USA) and have derived its empirical analytical equation (see Supplementary Material, File S15, eq. S15.13). We used that equation for all convolution operations. In order to perform a fast convolution of the spectrum with $P_{\lambda, instr}$, the Fast-Fourier Transform (FFT) convolution [44] was used instead of the direct integration in Eq. (22). Two C subroutine libraries for computing a discrete Fourier transform, FFTW [45] (ver. 3.3.5, freely available under GNU GPL) and Intel MKL [46] (ver. 2018.1.156, freely available for academic institutions) were implemented for the FFT and inverse FFT procedures.

3.3. Model vs. experiment spectra comparison technique

For the spectrum modeling itself, we set the central wavelength of the spectral range λ_C and determine its boundaries $[\lambda_{low}, \lambda_{upp}]$ by the following expressions:

$$\begin{cases} \lambda_{low} = \lambda_C - \Delta\lambda_{CCD}/2 - \Delta\lambda_{instr}/2 \\ \lambda_{upp} = \lambda_C + \Delta\lambda_{CCD}/2 + \Delta\lambda_{instr}/2 \end{cases} \quad (24)$$

where $\Delta\lambda_{CCD} = N_{CCD} w_{\lambda, pix}$ is the width of the observed spectral window (nm), $w_{\lambda, pix}$ is the spectral pixel width of ICCD (nm):

$$w_{\lambda, pix} = 10^{-3} w_{pix} \cdot FR \cdot RLD \quad (25)$$

$\Delta\lambda_{instr}$ (nm) is the effective part of the instrumental function ($P_{\lambda, instr} > \delta$, $\Delta\lambda_{instr} = 200 w_{\lambda, pix}$ for our experimental setup, see Supplementary Material, File S15, sections 3 and 4). Consideration of $\Delta\lambda_{instr}$ is necessary in order to correctly convolve spectra in the vicinity of the boundaries.

Generally, RLD and $w_{\lambda, pix}$ are nonlinear functions of wavelength (see Supplementary Material, File S15, eq. S15.7), while $w_{\lambda, pix}$ can be additionally disturbed by aberrations in the focal plane of the spectrograph within the observed spectral window. As a result, the experimental wavelength calibration function cannot be theoretically predicted with sufficient accuracy. Thus, for comparison of the model spectrum with the experimental one, we use in modeling the experimental wavelength bounds $\lambda_{low, exp}$ and $\lambda_{upp, exp}$ and neglect the nonlinear behavior of $w_{\lambda, pix}$:

$$\begin{cases} w_{\lambda, pix, exp} = (\lambda_{upp, exp} - \lambda_{low, exp}) / (N_{CCD} - 1) \\ \lambda_{low} = \lambda_{low, exp} - \Delta\lambda_{instr}/2 \\ \lambda_{upp} = \lambda_{low, exp} + \Delta\lambda_{instr}/2 \\ \lambda_C = (\lambda_{upp, exp} + \lambda_{low, exp}) / 2 \end{cases} \quad (26)$$

Experimental spectra were additionally interpolated by piecewise linear functions for a pixel-by-pixel (point-by-point) comparison with model spectra. Pearson's correlation coefficient (r) was served as a merit function of similarity between spectra (the model and experimental spectra are marked with the subscripts *mod* and *exp*, respectively):

$$r(T, n_e, m) = \frac{\sum_{i=1}^{N_{CCD}} (S_{mod, i} - \overline{S_{mod}})(S_{exp, i} - \overline{S_{exp}})}{\sqrt{\sum_{i=1}^{N_{CCD}} (S_{mod, i} - \overline{S_{mod}})^2 \cdot \sum_{i=1}^{N_{CCD}} (S_{exp, i} - \overline{S_{exp}})^2}} \quad (27)$$

3.4. Common details of spectra modeling

The presented algorithm for the plasma spectra modeling (Section 2) was implemented in the C++ language and compiled with Microsoft Visual Studio 2010. Our energy level (g_i and E_i , E_{ion}) and atomic line (A_{ij} , g_j , E_j , E_i , E_j) lists are composed of NIST [31] and Kurucz [47] data. A custom-made database powered by MySQL [48] contains approximately 500,000 lines of the elements with the known A values within the range 0.017–100,000 nm. Stark broadening parameters of the spectral lines are continually added from Stark-B database (calculated data) [49], Griem's calculated data [50], and numerous literature sources from the NIST bibliography database on atomic spectral line broadening (both experimental and calculated data) [51]. Since Stark parameters are unknown for the most of the lines, the modeling algorithm used “default” user-defined values for them. The necessary input parameters were loaded to the program from an *.ini file:

- plasma parameters (T , n_e , m , $\{c\}$);
- default Stark broadening parameters (width at the specified electron density, zero shift);
- database priority for A values (Kurucz's A values were used within the regions near 265 and 400 nm, and NIST A values were most suitable in our case within the range near 538 nm);
- spectrograph and detector parameters (see Table 1);
- number of initial calculated points per detector pixel (we used the initial wavelength step $\Delta\lambda = w_{\lambda, pix}/10$ for spectra calculation, see Section 2.4 and Supplementary Material, File S15, section 5);
- central wavelength of the range or experimental wavelength range boundaries (see Section 3.3);
- accuracy δ (the value 10^{-5} was used, see Section 2.4);
- lines identification cut-off criteria ($\alpha_{peak, exp}$, $\alpha_{line, user}$, see Section 4.5).

To assess the capabilities of our algorithm, we investigated plasma evolution, modeling performance and behavior of merit function (Eq.

(27)) within three spectral regions centered at 265 nm, 400 nm and 538 nm. Emission spectra were modeled with the following plasma parameters: $\{T \in [0.5; 1.5], \text{ step } 0.02 \text{ eV}; \log_{10}(n_e, \text{ cm}^{-3}) \in [16.0; 17.9], \text{ step } 0.02; m \in \{0.5, 1, 2, 5, 10, 20, 50, 100, 200\} \text{ ng}\}$.

4. Results and discussion

4.1. Stark parameters

Based on extensive measurements of Stark widths of numerous Fe II lines within the range 230–300 nm in laser-induced plasma by Aragon and Aguilera [52,53] (typical values of 4–6 pm at $n_e = 10^{17} \text{ cm}^{-3}$, $T \approx 12,900\text{--}15,200 \text{ K}$), we set the default w_{Stark} equal to 6 pm at $n_e = 10^{17} \text{ cm}^{-3}$ within the range 255–276 nm. Although we have measured Stark widths of a few Mn I lines within 390–410 nm [34], to our best knowledge, there is no data on Stark parameters of iron lines within this spectral range. Therefore, we estimated the default value of w_{Stark} (12 pm at $n_e = 10^{17} \text{ cm}^{-3}$) as the one providing the best fit of the experimental spectra at the n_e value independently measured using the Fe I 538.34 nm and H α 656.28 nm lines.

A number of Stark parameters of Fe I lines within 529–547 nm belonging to 3 multiplets ($z^5G^\circ\text{--}e^5H$, $z^5D^\circ\text{--}e^5D$, $z^3G^\circ\text{--}e^3H$) were compiled by Konjević et al. at different T values in the review [37]. We used these data with the exception of 5404 Å line (probably erroneously attributed in the review to the multiplet $z^5G^\circ\text{--}f^6G$ due to wavelengths closeness of the Fe I 540.415 and Fe I 540.412 lines from the multiplets $z^3G^\circ\text{--}e^3H$ and $z^3G^\circ\text{--}f^6G$, respectively [31]). We assumed the assignment of the 5404 Å line to the $z^3G^\circ\text{--}e^3H$ multiplet in accordance with the original experimental paper by Lesage et al. [54]. The same values of w_{Stark} were implemented for each component of a multiplet neglecting the difference between them at the experimental uncertainty level (15–30%). We also neglected the temperature dependence of w_{Stark} [54] and chose the values obtained at the temperature closest to our measurements. For example, w_{Stark} was taken equal to 27.63 pm at $n_e = 1.9 \cdot 10^{16} \text{ cm}^{-3}$ and $T = 8697 \text{ K} \approx 0.75 \text{ eV}$ (i.e. the Stark width of Fe I 538.337 nm) for all components of the $z^5G^\circ\text{--}e^5H$ multiplet. We estimated several Stark broadening parameters from the spectra of low-alloyed steel UG 112 registered at different delay times by plotting the parameter value vs electron density (n_e was determined from the Fe I 538.337 nm line Stark broadening). Stark parameters of all Fe I lines within the window 529–547 nm used during the modeling procedure are collected in a supplementary data file (File S1); the default w_{Stark} was set to 36 pm at $n_e = 10^{17} \text{ cm}^{-3}$ here.

4.2. Calculations performance

A spectrum calculation time, generally, depends on the number of involved lines, their FWHM (mainly, Stark width, which in turn depends on electron density), the number of points per pixel (or wavelength step) and accuracy δ . We have performed the following test: modeling of 5000 spectra (single thread application) for the CRM 475 sample (Tables 2, 14 elements) at $T = 0.74 \text{ eV}$, $\log_{10}(n_e, \text{ cm}^{-3}) = 16.76$, $m = 40 \text{ ng}$, initial spectral range 390.90–413.86 nm with: (i) 16,206 points (10 points per pixel, wavelength step 1.417 pm) and (ii) 8102 points (5 points per pixel, step 2.834 nm) and default Stark width of 0.012 nm at $n_e = 10^{17} \text{ cm}^{-3}$. The final spectral range after convolution with P_{instr} was 392.53–412.23 nm (13,900 and 6950 final points for the cases (i) and (ii), respectively). Out of 5825 lines found in the database, 1661 were involved in the spectra calculations according to the cutoff criterion $\delta = 10^{-5}$ (Eq. (15)). Default line profiles for each of the 14 elements with $w_L = w_{\text{Stark,FWHM}} = 0.0069 \text{ nm}$ (see Section 2.4) and 6 specific line profiles for Mn I lines with the known Stark widths [34] were calculated. The calculation times per one spectrum are presented in Table 3.

The wavelength step of 1.417 pm (10 points per pixel) corresponded to ~ 6 points per FWHM of the default Voigt profile of an iron line with

Table 3

Calculation performance per one spectrum of the high-alloy steel CRM 475 (Table 2) within the range 390.90–413.86 nm.

Number of points ^a	Wavelength step, pm	Calculation time per one spectrum, ms		
		“Old” PC ^b , FFTW [45]	“Modern” PC ^c , FFTW [45]	“Modern” PC ^c , Intel MKL [46]
16,206	1.417	42	14.6	6.73
8102	2.834	15	5.44	3.31

^a Total number of initial calculation points in the spectrum before convolution with the instrumental profile and before wavelength discretization (see Section 2.6).

^b AMD Athlon 64 $\times$ 2 6000+ (Windsor) processor, 3.0 GHz, released in May 2006, DDR2–800 memory, OS Windows XP.

^c Intel Core i7 6700 (Skylake) processor, up to 4.0 GHz, released in Q3’2015, DDR4–2133 memory, OS Windows 7.

$w_L = 6.9 \text{ pm}$, $w_G = w_{\text{Doppler}} = 3.6 \text{ pm}$. However, a reduced precision (5 points per pixel, ~ 3 points per FWHM) provides the same quality of convoluted spectra at the above-mentioned plasma parameters: the mean relative difference between the final spectra was 0.6% (maximum – 2.0%), while the calculation time was reduced approximately by half in the case of a “modern” PC with the Intel MKL library [46], and by a factor of 2.7 in the case of the FFTW library [45] (Table 3). Thus, an appropriate number of points per FWHM of the narrowest Voigt line profile should be chosen by the user for maximal performance.

4.3. Behavior of the cost function $r(T, n_e, m)$

First of all, we have investigated the behavior of the modified correlation function f (this representation of Pearson's r was done for clarity of figures presentation):

$$f(T, n_e, m) = \frac{1 + r(T, n_e, m)}{1 - r(T, n_e, m)} \quad (28)$$

in the space of the model plasma T and n_e at a fixed plasma mass $m = 20 \text{ ng}$. We have simulated a set of the model spectra (3744 spectra in total) of the high-alloy steel (Table 2) within the region 392.5–412.2 nm with T ranging from 0.3 to 2.0 eV with a step of 0.05 eV and $\log_{10}(n_e, \text{ cm}^{-3})$ ranging from 14.5 to 18.3 with a step of 0.1. The calculation grid steps were reduced near the f maximum ($T \in [0.56; 1.32]$; $\log_{10}(n_e, \text{ cm}^{-3}) \in [16.5; 17.7]$) to 0.02 eV and 0.02, respectively. The correlation function f for the spectrum measured at the delay of 4 μs (gate 250 ns) had only one maximum at $T = 0.72 \text{ eV}$, $\log_{10}(n_e) = 16.80$ ($f(T, \log_{10}(n_e)) = 135.19$, $r = 0.98531$, Fig. 2). Despite the clear f maximum the experimental spectrum correlates well with the model spectra ($f > 100$, $r > 0.9800$, orange and red fields in Fig. 2) within relatively large ranges of T and $\log_{10}(n_e)$: 0.66–0.80 eV and 16.45–17.10 (4.5 times difference in terms of n_e), respectively. The possible reasons are the large number of observed iron lines with close upper-level energies and strong self-absorption of the lines. A weak f dependence on n_e can be related with the absence of strongly Stark-broadened lines within the spectral region. If the Stark and Doppler widths are several times or more smaller than the instrumental width, and the line is weakly self-absorbed, the experimental line profile will be mainly given by the instrumental profile. On the contrary, self-absorbed line profiles can be more sensitive to n_e changes even at small w_{Stark} values in comparison with the instrumental width (see Eqs. (21), (8) and (14)), but the Stark width should be known for the correct spectrum calculation.

Secondly, we have investigated the behavior of the correlation maximum $f(T_{\text{best}}, \log_{10}n_{e,\text{best}})$ vs. plasma mass in two spectral regions (265 and 400 nm, Fig. 3). 4896 model spectra for each mass (see Section 3.4 for the details) were calculated and correlated with the

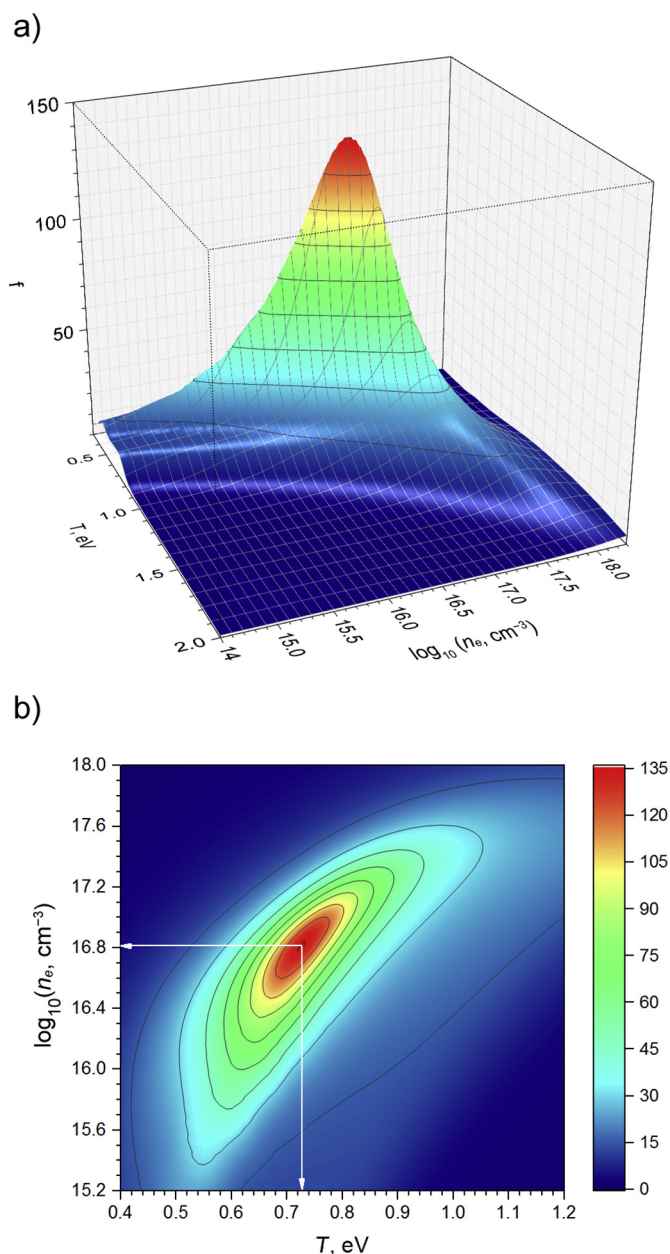


Fig. 2. A 3D colormap (a) and its central part projection (b) of the correlation parameter $f(T, \log_{10}n_e)$ (Eq. (28)) between the model and experimental spectra of the high alloy steel CRM 475 (Table 2) at the delay 4 μ s (gate 250 ns) for the fixed model plasma mass (20 ng) within the wavelength range 392.5–412.2 nm. The best fitted T and $\log_{10}n_e$ are indicated on the panel (b).

experimental spectra at different delays after the laser pulse. The model and experimental data correlate well at short delay times (1–2 μ s) within the range near 265 nm (Fig. 3a). Medium f values are observed at medium delays, and the correlation is poor at long delays (7–10 μ s). A clearly observed maximum corresponding to the best fitted mass of 5 ng at the delays of 1–2 μ s becomes less pronounced at later delays. Such a behavior can be explained by the presence of the both atomic (Fe I) and ionic lines (Fe II, Cr II) within the region 255–276 nm and inhomogeneity of the plasma. The emission of the ions and atoms can originate from different parts of the plasma plume characterized by their own temperature (see Section 4.4): the ionic line emission mainly comes from the “hot” part of the plasma, and atomic Fe I lines originate in the “cold” part. The plasma morphology at short delays resulted in prevalence of the ionic iron lines in spatially integrated spectra within

the range 255–276 nm, while at longer delays the contribution of both species to the emission spectra became comparable. Therefore, the spectra in the UV range cannot be fitted well by the uniform plasma model at late stages of the plasma lifetime (the decrease of the correlation strength is shown in Fig. 4). The plasma profiles obtained from an Abel inversion of spatially and temporally resolved spectra of an iron plasma in air [12] confirm our assumption: the total ionic species number in the plasma core exceeds the atomic one at delays up to 5–6 microseconds.

On the contrary, the model spectrum did not fit well to the experimental one within the range near 400 nm at the delay of 1 μ s (Fig. 3b), while the f -value was high at longer delays (Fig. 4) with a prominent maximum corresponding to the mass of 10–20 ng (Fig. 3b). The relatively poor correlation ($r=0.979$) at the shortest delay can be explained by the imprecise modeling of iron line profiles at relatively high n_e (see Section 4.4) due to the absence of data on the Stark parameters for any Fe I lines within the range (the same “default” value was used instead). The spectra at all delays were well explained within the assumption of a uniform plasma since only atomic lines mainly originated from the “cold” plasma periphery were observed within the region.

Within the range 529–547 nm the absorption coefficient (see Eqs. (20) and (8)) for most of the observed Fe I lines was small in comparison with the lines from the range near 400 nm due to lower transition probabilities, higher excitation potentials, and higher Stark widths. As a consequence, the spectra were practically insensitive to the plasma mass (and optical path length, see Eqs. (16) and (17)) variations within the considered grid values. Thereby, we decided to use the mid best fitted plasma mass obtained in the region 400 nm for spectra fitting in the 538 nm region (namely, 20 ng for all delays). The range 529–547 nm was dominated by Fe I lines; therefore, the trend of the correlation function was close to that of the 400 nm region (Fig. 4). Note that the region 529.3–533.3 nm was excluded from the correlation procedure in the case of highly-alloyed steel due to strong discrepancies of several Cr I lines in the model spectrum with respect to the experimental one (see Section 4.6). The f -value was poor ($f=50$, $r=0.96$, Fig. 4) at a short delay of 2 μ s, while the measured data at 1 μ s cannot be adequately fitted due to strong influence of the uncertainties of the Stark parameters on the model results.

It should be noted that the plasma mass in the present model is directly related to the plasma volume V and optical path length l (see Eqs. (16) and (17)). Therefore, the lines with large absorption coefficient $\kappa(\lambda)$ (and large ϵ_{λ} , respectively) are more sensitive to changes in the plasma mass (see Eqs. (19)–(21)). If absorption is very small ($\kappa_{exp}(\lambda) \cdot l_{exp} \ll 1$, $\kappa_{exp}(\lambda)$ – experimental absorption coefficient, l_{exp} – experimental plasma size along the line of observation) for all of the observed lines within the specific spectral region, the best-fit plasma mass cannot be found due to the spectrum profile insensitivity to the plasma mass changes. We can evaluate the critical mass value m_{crit} only from the relation $\kappa(\lambda) \cdot l_{crit} \approx 0.01$, when the self-absorption begins to distort the model spectrum profile.

To sum up the discussions, we should simultaneously observe the lines with significantly different upper level energies, self-absorbed lines, and lines with well-known Stark parameters meeting the criterion $w_{Stark} \geq w_{Instr}$ to clearly fit the parameters T , n_e and m .

4.4. Evolution of plasma parameters

We have investigated the plasma electron density and temperature of the high-alloy steel at the different delays after the laser pulse. In addition to the parameters of the best fit model spectra we have measured electron density by Stark broadening of Fe I 538.33 nm and Balmer H α 656.28 nm lines (hydrogen is admixed to the plasma from the surrounded air [55]). The lines were approximated by Lorentzian profile and the Stark width was obtained by subtraction of instrumental width from the fitted line width: $w_{Stark} = w_{L,exper} - w_{Instr}$. We have

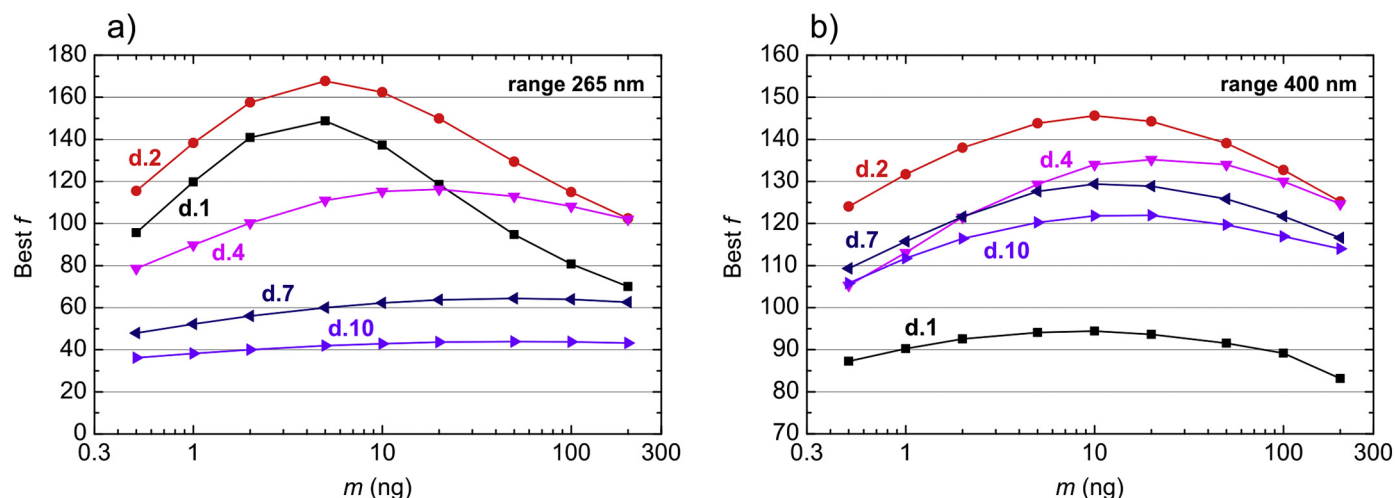


Fig. 3. The best correlation parameter $f(T, \log_{10}n_e)$ between the model and experimental spectra of the high-alloy steel CRM 475 (Table 2) obtained at different delays after the laser pulse (μ s) as a function of the model plasma mass m (ng, logarithmic scale).

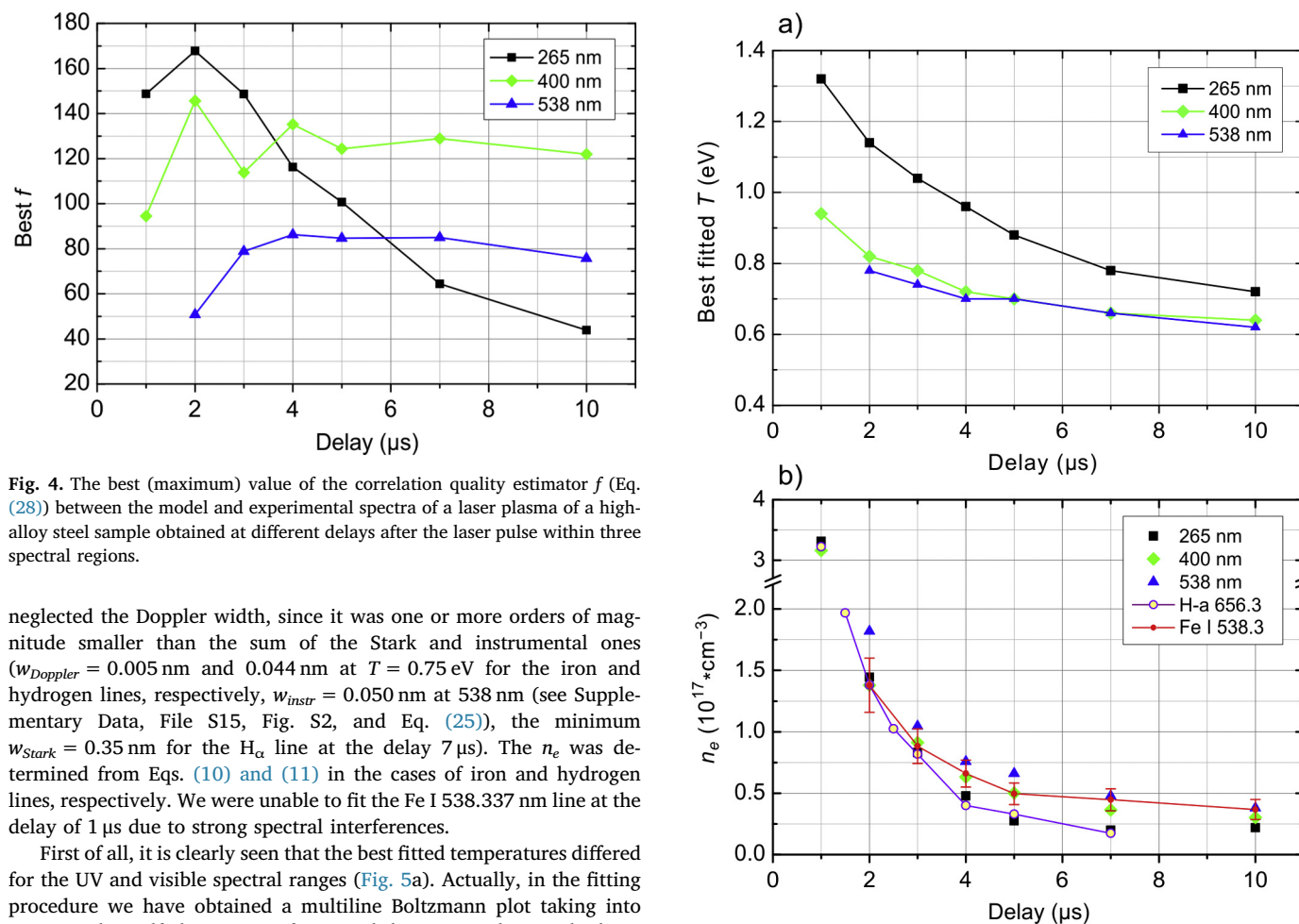


Fig. 4. The best (maximum) value of the correlation quality estimator f (Eq. (28)) between the model and experimental spectra of a laser plasma of a high-alloy steel sample obtained at different delays after the laser pulse within three spectral regions.

neglected the Doppler width, since it was one or more orders of magnitude smaller than the sum of the Stark and instrumental ones ($w_{\text{Doppler}} = 0.005$ nm and 0.044 nm at $T = 0.75$ eV for the iron and hydrogen lines, respectively, $w_{\text{instr}} = 0.050$ nm at 538 nm (see Supplementary Data, File S15, Fig. S2, and Eq. (25)), the minimum $w_{\text{Stark}} = 0.35$ nm for the H_{α} line at the delay 7μ s). The n_e was determined from Eqs. (10) and (11) in the cases of iron and hydrogen lines, respectively. We were unable to fit the Fe I 538.337 nm line at the delay of 1μ s due to strong spectral interferences.

First of all, it is clearly seen that the best fitted temperatures differed for the UV and visible spectral ranges (Fig. 5a). Actually, in the fitting procedure we have obtained a multiline Boltzmann plot taking into account the self-absorption of spectral lines. We discussed above (Section 4.3) that in the case of spatially integrated measurements the ionic Fe II and Cr II lines emitted by the “hot” plasma core mainly contributed to the spectra within the range 255–276 nm, while the spectra in the visible regions generally consisted of atomic lines emitted by the “cold” plasma periphery. The relative difference between the temperatures of the “hot” and “cold” plasma parts gradually decreases from 40% (with respect to the lower temperature) at the delays of (1–2) μ s to 12% at 10μ s. Our results are in good agreement with the difference between temperatures of a Fe–Ni plasma determined with the use

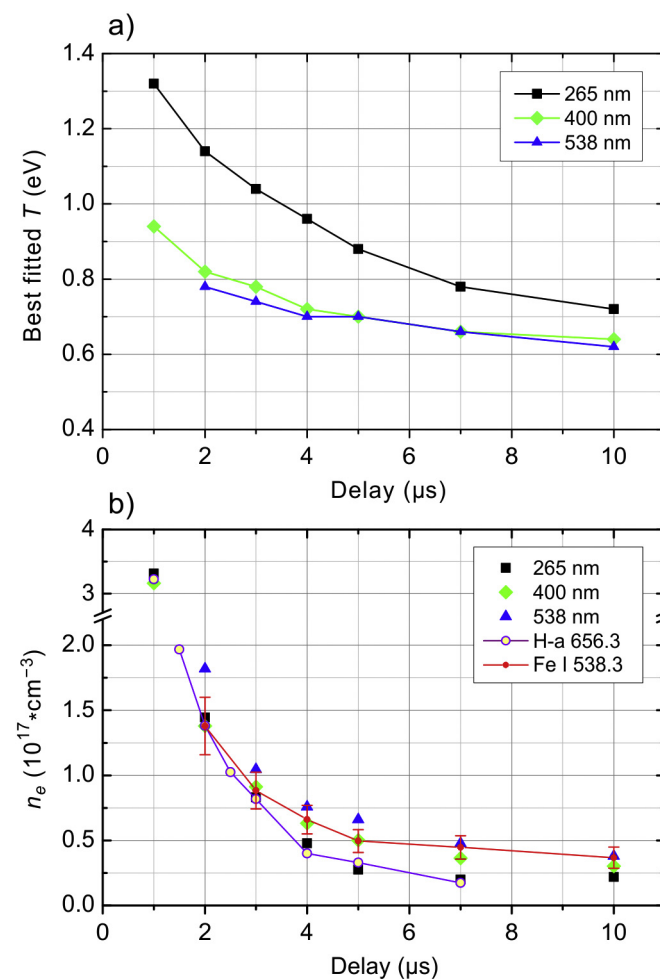


Fig. 5. Best fitted temperature (a) and fitted and measured electron density (b) for experimental spectra of a high-alloy steel in three spectral regions. The error bars on the electron density plot (b) correspond to the uncertainty of the Stark width parameter for the Fe I 538.337 nm line (15% [37,54]).

of Fe I and Fe II lines in the case of spatially integrated measurements by Aguilera and Aragón [56]. Additionally made spatially resolved measurements demonstrated [56] that the emission of the Fe II lines originates from the inner “hot” zone of the plume near the plasma axis

and close to the sample surface, while the Fe I lines are emitted mainly from the outer part (“cold” zone) of the plasma plume widely distributed in the direction perpendicular to the plasma axis.

Generally, the values of electron density (Fig. 5b) obtained from the spectra fitting procedure as well as from different spectral lines and species vary less than the temperature values (Fig. 5a). We state this fact since the sensitivity of line intensities to temperature is an order of magnitude or more greater than sensitivity to electron density. This is because the number density determined by either Saha Eq. (1) or Boltzmann Eq. (6) depends on T exponentially rather than linear on n_e . The best fitted electron densities at the delay of 1 μ s obtained in the regions near 265 and 400 nm are very close to the one determined from the H_α line profile. The n_e values for the range near 538 nm are slightly greater than the others due to strong broadening and shifts of numerous observed Fe I lines and large uncertainties of Stark parameters (correlation with experimental spectra was relatively poor). The n_e values determined from the H_α and Fe I lines differed starting from the delay of 4 μ s. It should be noted that the best fitted n_e in the visible regions was close to the value obtained from the Fe I line, while the fitted n_e value in the UV region was close to the one from the H_α line. We assume that the Fe II and H_α emission originated from the same “hot” zone of the plasma (H_α has a high excitation potential of the upper level – 12.087 eV [31]), while the emission of the Fe I lines originated from the “cold” zone. Again Aragón and Aguilera have studied the electron density deduced from the H_α 656.28 and Fe I 538.337 lines [55] in spatially resolved and spatially integrated measurements of a ferrosilicon plasma (atomic fractions Fe:Si = 4:6; delay 2.85 μ s after a 100 mJ 1064 nm Nd:YAG laser pulse; gate 0.3 μ s). They showed that emission of these lines originates from different zones of the plasma plume. Axially integrated n_e obtained from the H_α line was 1.2 times greater than the one deduced from the Fe I line. However, it should be noted that the electron impact broadening parameter used in Ref. [55] for the Fe I line was 1.3 times greater ($w_{\text{Stark}}/n_e = 0.194 \cdot 10^{-17} \text{ nm} \cdot \text{cm}^3$ [57]) than the one used in our calculations ($0.145 \cdot 10^{-17} \text{ nm} \cdot \text{cm}^3$ [54,37]). Considering this fact, we have a very good agreement with the results of Aragón and Aguilera, i.e. the close values of n_e in spatially integrated measurements from the H_α vs. Fe I 538.337 nm lines at the delays of 2 and 3 μ s: $(1.38 \pm 0.01) \times 10^{17} \text{ cm}^{-3}$ vs. $(1.38 \pm 0.22) \times 10^{17} \text{ cm}^{-3}$ and $(0.82 \pm 0.01) \times 10^{17} \text{ cm}^{-3}$ vs. $(0.88 \pm 0.14) \times 10^{17} \text{ cm}^{-3}$, respectively).

Finally, we deduced the best fitted values of T and n_e at a fixed plasma mass of 20 ng from our dataset (Section 4.3) and calculated the optical path length l (Eq. (16)). The observed curves can be related to plasma expansion (Fig. 6), and we can conclude that the stationary

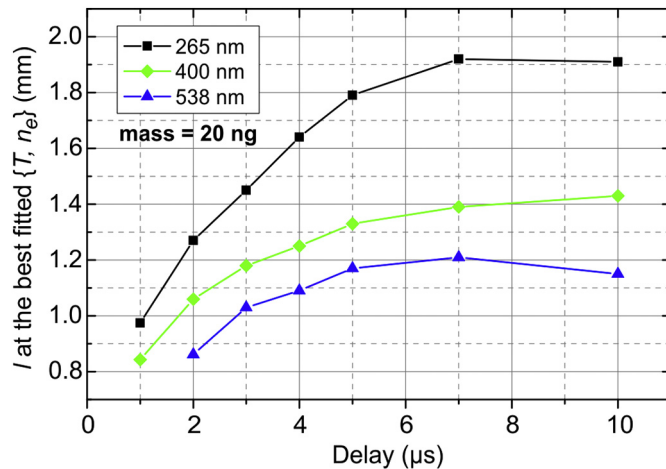


Fig. 6. Calculated optical path length (Eq. (16)) for the best fitted T and n_e at a fixed plasma mass of 20 ng for high-alloy steel spectra within three spectral ranges.

model provides a physically meaningful description of plasma evolution.

4.5. Line identification algorithm

One of our main goals was to utilize the spectral modeling described above for automated line identification in experimental spectra and selection of lines most suitable for diagnostics and analytical application. The identification procedure was previously proposed by us [1] without consideration of self-absorption. A brief description of the modified approach is given below. Assuming the elemental composition of the plasma to be known, we retrieve the plasma temperature, electron density and mass from the best fitted model spectrum (see Section 4.3). In the present work we use a regular grid of the parameters $\{T, n_e, m\}$ (see Section 3.4), but an optimization algorithm can be used to minimize the function $1 - r(T, n_e, m)$ (Eq. (27)). In the best fitted spectrum we define a “peak” as the spectral range between two adjacent local minima with coordinates $\lambda_{\min, i}$ and $\lambda_{\min, i+1}$ containing a local maximum at a wavelength $\lambda_{\text{peak}, k}$ (Fig. 7). Spectral lines are attributed (or “linked”) to those peaks.

To consider an extreme case in the peak finding algorithm (spectral region with a plateau) we find a local maximum or a “left”/“right” local minimum (shorter/greater wavelengths towards the closest local maximum) according to the criterion of the first derivative:

$$\begin{cases} \begin{cases} S_i - S_{i-1} > 0 \\ S_{i+1} - S_i \leq 0 \end{cases} \Rightarrow \text{local maximum} \\ \begin{cases} S_i - S_{i-1} < 0 \\ S_{i+1} - S_i \geq 0 \end{cases} \Rightarrow \text{“right” local minimum} \\ \begin{cases} S_i - S_{i-1} = 0 \\ S_{i+1} - S_i > 0 \end{cases} \Rightarrow \text{“left” local minimum} \end{cases} \quad (29)$$

This allows selection of the sole point within the plateau as a local extremum.

The height of the k -th peak H_k in the model spectrum was calculated as:

$$H_k = S_{\max, k, \text{mod}} - 0.5 |S_{\min, i, \text{mod}} - S_{\min, i+1, \text{mod}}| \quad (30)$$

Weak peaks are omitted using the following criterion:

$$\frac{H_k}{S_{\max, \text{mod}}} < \frac{3s_{b, \text{exp}}}{S_{\max, \text{exp}}} = \alpha_{\text{peak, exp}} \quad (31)$$

where $S_{\max, \text{mod}}$ and $S_{\max, \text{exp}}$ are the maximal intensities within the considered model and experimental spectra, respectively, $s_{b, \text{exp}}$ is a noise (standard deviation of the background signal calculated from the range free of spectral lines in the experimental spectrum). All lines within the peak area may be attributed to it. To reduce the number of lines, we apply the following criterion that selects only usable lines:

$$\frac{I_{\lambda_{c,j}, \text{line}}}{\max_j I_{\lambda_{c,j}, \text{line}}} \geq \alpha_{\text{line, user}}, \quad \lambda_{c,j} \in [\lambda_{\min, i}, \lambda_{\min, i+1}] \quad (32)$$

where α_{line} is a cut-off level defined by the user and $I_{\lambda_{c,j}, \text{line}}$ is the intensity of the j -th line within the peak area at the central wavelength defined by Eqs. (19) and (20) at $\lambda = \lambda_{c,j}$ (wavelength indexes c, j are omitted below for clarity):

$$I_{\lambda, \text{line}}(l) = \frac{\varepsilon_{\lambda, \text{line}}}{(\varepsilon_{\lambda}/B_{\lambda})} (1 - e^{-(\varepsilon_{\lambda}/B_{\lambda})l}) \quad (33)$$

$\varepsilon_{\lambda, \text{line}}$ is retrieved from Eq. (8) for the specific transition (k, z, i) at $\lambda = \lambda_{c,j}$. The contribution (“weight”) w_j of the line intensity to the observed intensity is estimated as:

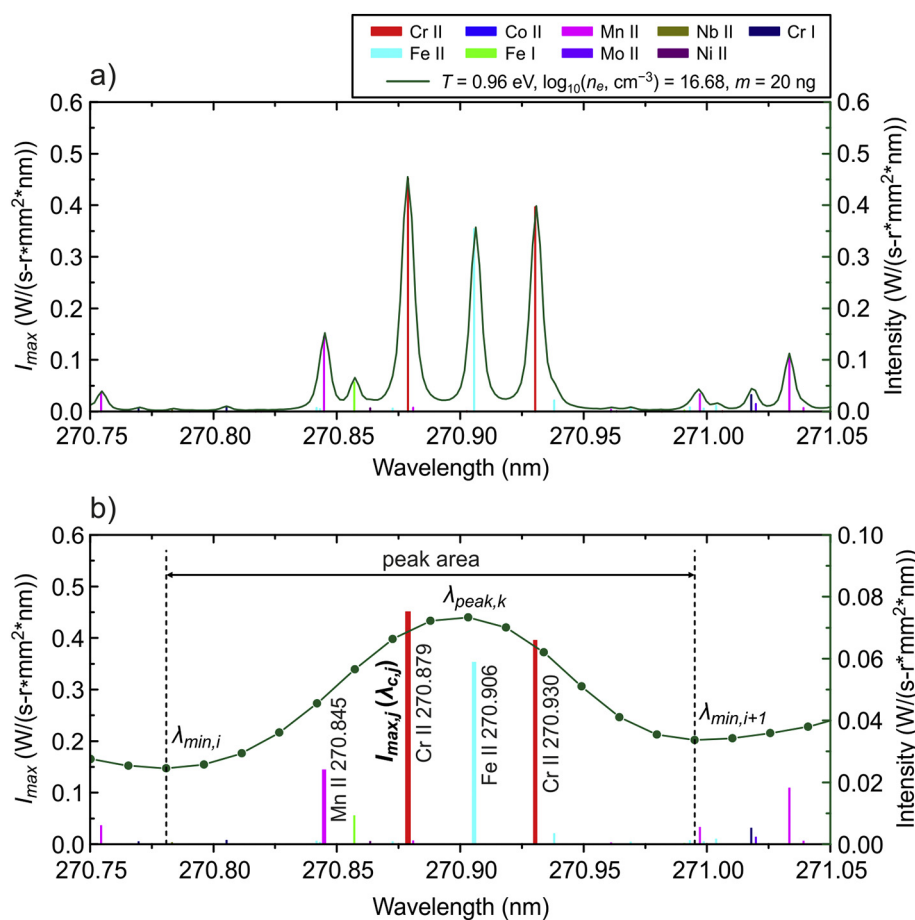


Fig. 7. (a) A model spectrum of a high-alloy steel within a narrow spectral region before convolution with the instrumental function. Plasma parameters are given in the legend. (b) The same spectrum after convolution with the instrumental function (see Section 3.2 and Supplementary Data, File S15, sections 3–5): attribution of the lines to the observed peak is illustrated.

$$\omega_j = I_{\lambda_{c,j},line} / \sum_j I_{\lambda_{c,j},line} \quad (34)$$

We also calculate line emission transmittance T_λ at $\lambda = \lambda_{c,j}$ to estimate self-absorption:

$$T_\lambda = \frac{I_{\lambda_{c,j},line}}{\varepsilon_{\lambda_{c,j},line} l} \quad (35)$$

Even though the model spectrum used for identification is well correlated with the experimental one, we can encounter two types of possible misidentification in the suggested approach:

1. The peak is absent in experiment spectrum but present in the model one. It can be related to erroneous atomic data (transition probability, multiplet assignment, etc.), or to the absence of the impact parameters for strongly Stark-broadened lines in the database, or to the influence of the lowering factor on levels close to the ionization potential.
2. The peak is absent in the model spectrum but present in the experiment one. It can be related to missing data in databases (unknown transition or incomplete/incorrect transition parameters).

4.6. Best fitted spectra and lines analytical performance

For line identification in the measured spectra, the threshold parameter α_{line} (Eq. (32)) was set at the level 0.2, while the peak identification threshold α_{peak} (Eq. (31)) was deduced from the averaged spectra (the standard deviation of the intensity within the spectral range free of spectral lines was served as s_b). The identification results for both the high- and low-alloy steels (Table 2) spectra measured at the delay of 4 μ s are summarized in Table 4, while the measured and best fitted

spectra of the high-alloy steel are presented in Fig. 8. The spectra and complete lists of identified lines obtained by the procedure described in Section 4.5 for both the high- and low-alloy steels can be found in supplementary materials (Files S2–S7 for the high-alloy steel and S9–S14 for the low-alloy steel).

Firstly, we should note that model adequacy in relation to the state of real plasma is confirmed by good agreement of the best fitted parameters with the results of plasma diagnostics by Aragón and Aguilera [12,55,56] for the Fe alloys at close experimental conditions (see Sections 4.3 and 4.4). Secondly, the validity of our model is supported by closeness of the best-fitted parameters for high- and low-alloy steels: the best-fitted temperatures in UV range were close to 1.00 eV and to 0.72 eV in the visible ranges, while the values of the electron density were of about $(6-8) \times 10^{16} \text{ cm}^{-3}$ in all ranges. Nevertheless, we note a few discrepancies between the experimental and model spectra. To avoid redundant discussion we focus on the spectra of the high-alloy steel (Fig. 8) as the most complicated case.

The Cr I lines from the multiplet $3d^5(^4G)4s\ ^5G - 3d^5(^4G)4p\ z\ ^3H^\circ$ near 402.5 nm are stronger in the model spectrum in comparison with experiment. The transition probabilities are absent in the NIST database [31] for these lines and were retrieved from Kurucz's database [47] (reference "K88", semiempirical calculations by Kurucz [58]). Also, we observed the resonance line of Ca II at 393.366 nm, which can be explained by presence of Ca traces in the steel (Ca was not certified).

Another discrepancy was observed within the range 529.3–533.3 nm containing Cr I lines from the multiplet $3d^5(^6S)4p\ z\ ^7P^\circ - 3d^5(^6S)4d\ e\ ^7D$. When the lines are included into the calculations, the intensities of the peaks in the model spectrum near 529.8 and 532.9 nm are much higher than experimental ones (dashed line in Fig. 8c), while when we exclude the lines of this multiplet, the model intensities are lower than experimental ones (solid blue line in Fig. 8c). We

Table 4Results of the automatic lines identification in the steels spectra at the delay 4 μ s.

Spectral range ^a , nm	Spectral range ^b , nm	<i>T</i> , eV	$\log_{10}(n_e, \text{cm}^{-3})$	<i>m</i> , ng	<i>l</i> , mm	$3s_b/I_{\max}$	<i>nldb</i>	<i>nlc</i>	<i>np</i>	<i>nli</i>
High-alloy steel CRM 475										
253.336–278.110	255.094–276.352	0.96	16.68	20	1.637	0.005	11,700	2276	106	275
390.894–413.863	392.535–412.221	0.72	16.80	20	1.254	0.009	5825	1645	79	152
527.806–548.443	529.29–546.96	0.70	16.88	20	1.088	0.004	3241	871	50	73
Low-alloy steel UG112										
253.336–278.111	255.094–276.353	1.04	16.90	20	1.390	0.010	12,642	1760	89	228
390.925–413.891	392.566–412.250	0.76	16.84	20	1.290	0.005	6789	977	78	133
527.807–548.434	529.29–546.95	0.70	16.88	20	1.048	0.004	3902	612	45	58

T, $\log_{10}(n_e, \text{cm}^{-3})$, *m* – best fitted plasma parameters; *l* – calculated plasma length (Eq. (16)); $3s_b/I_{\max}$ – threshold for peaks recognition (Eq. (31)) deduced from experimental noise (see text); *nldb* – total number of lines selected from the database for the specified elements within the spectral range [1]; *nlc* – the number of lines involved into spectra calculations at the threshold $\delta = 10^{-5}$ (see Eq. (15)); *np* – the number of identified peaks within the observed spectral region; *nli* – the number of lines linked to the peaks at the threshold $\alpha_{\text{line}} = 0.2$ (Eq. (32)).

^a Spectral range involved in calculations including instrumental function width (Eq. (26)).

^b Observed spectral range.

summarized the information about the transition probabilities and references for the multiplet in the supplementary table (File S8). Although some literature is not available in public sources (for example, private communication by J. M. Bridges, see comments in Supplementary S8), it can be concluded that transition probabilities from the NIST and Kurucz databases are the same or very close to each other. We assume that the discrepancies between the model and experimental intensities (spectrum profiles) are caused by significant differences of the used “default” Stark width from real ones for the Cr I multiplet $3d^5(^6S)4p\ z\ ^7P^\circ - 3d^5(^6S)4d\ e\ ^7D$. It is likely that the Stark widths are much greater than we assumed in the calculation. The latter conclusion is consistent with appearance of high pseudo-background (resulting from overlapping of strongly broadened lines) in the vicinity of these lines (Fig. 8c, red solid curve). Since the discussed multiplet overlaps with another Cr I multiplet $3d^44s^2\ a\ ^5D - 3d^5(^6S)4p\ z\ ^5P^\circ$ (two peaks at 529.669, 529.827 nm were clearly visible in the experimental spectrum in Fig. 8c) and Fe I lines at 532.804 and 532.853 nm, we exclude the six lines from the Cr I $3d^5(^6S)4p\ z\ ^7P^\circ - 3d^5(^6S)4d\ e\ ^7D$ multiplet from the final list of identified lines (Supplementary Data, File S7) due to its unknown contribution to the peak intensities.

Ionic iron line Fe II 531.661 was not fitted, as expected, by the model of homogeneous plasma, since the ions mostly emit from the “hot” ($T = 0.96$ eV at the delay 4 μ s) core of the plasma as discussed in details in Sections 4.3 and -4.4, while the measured spectrum within 529–547 nm region (and as a result the model one) consists mainly of atomic emission lines (Fe I, Cr I) originating from the “cold” ($T = 0.70$ eV) plasma zone.

Finally, let us consider the contribution of a transition to the peak intensity (Eq. (34)) as well as the degree of self-absorption (calculated as a transmittance at $\lambda = \lambda_c$ at the best-fitted parameters, Eq. (35)). Most significantly, the values are calculated without any preliminary characterization of the laser-induced plasma, which allows an assessment of the analytical performance of a particular line or its suitability for plasma diagnostics. The latter is important for calibration-free LIBS (or for monitoring plasma temperature in diagnostic application), since self-absorption, especially in lines of the major element, is perhaps the main problem disturbing accuracy. Indeed, the modeling results in only one isolated (weight > 0.9) iron line with low level of self-absorption (transmittance > 90%) in the 255–276 nm range, 6 such lines in the 392–412 nm range, and 12 in the 529–546 nm range. Considering the noise level of the experimental spectrum, only 5 lines from the latter range can be measured reliably. The greater number of potentially suitable lines in the green range is easily understandable due to growth of the number of lines with high energy of lower level as well as the resolving power of the Czerny–Turner spectrometer.

The model spectra are extremely helpful for proper analytical line selection. Since self-absorption gradually changes with element

content, the line list produced by the algorithm is extremely useful to select lines for a particular sample. During the 2nd LIBS Inter-Laboratory Comparison, chromium was determined in cast irons. Its content varied from 0.05 to 1 mass.% in the reference samples. We have preliminarily assessed that the Cr lines intensities in unknown sample spectra are lower or comparable with ones of the reference sample with 0.303 mass.% of Cr. Therefore, the best-fitted model spectrum for the sample with 0.303 mass.% Cr was found, and calculations in a wide spectral range (212–530 nm) with the best-fitted parameters demonstrated that the Cr I 425.435 and 427.48 nm lines are simultaneously: (i) almost free of spectral interferences (weight > 0.9), (ii) the strongest Cr lines, (iii) slightly self-absorbed (transmittance was about 0.95 despite the transition to ground state). We choose Cr I 427.48 nm line for analysis due to its slightly greater weight and transmittance values. Our conclusion that this line is ideal for analytical measurements was later confirmed by the high ranking of our results among participating laboratories. It should be noted that this line becomes not optimal for the samples with $c(\text{Cr}) \sim 1$ mass.% (calculated transmittance goes down to 0.88), and can hardly be used for high-alloy steel (> 10 mass.% of Cr, calculated transmittance becomes lower than 0.4).

Yet another example shows how a particular line can be proven unusable for analysis. Carbon is one of the “hard” elements for optical emission spectroscopy due to a small number of intense lines in near-UV–Vis range. The frequently-used C I 247.856 nm line is overlapped with ionic and, partially, atomic iron lines (Fe II 247.857 nm and Fe I 247.978 nm, respectively). Interferences with these lines with having very different excitation potential precludes existence of optimal plasma parameters for carbon determination in steels, which can be clearly demonstrated by the modeling. The lower bound of the resulting interference can be considered as the sum of the intensities of the ionic line at the central wavelength and the intensity in the wing of the Fe I 247.978 nm line resulted from the broadening effects. Neglecting the self-absorption, we can conclude from Fig. 9 that the ratio of the C I 247.856 nm line intensity ($c(\text{C})$) was set to 0.186 mass.% here) to the interfering ones ($I(\text{Fe II}) + I(\text{Fe I}) \cdot 0.0068$, assuming a Lorentzian instrumental function with FWHM = 0.02 nm) is extremely low at typical electron densities in laser-induced plasma. A detailed discussion and comparison with experimental data can be found in Ref. [59]. Thus, if the C I 193.09 nm line (vacuum wavelength) cannot be detected by a particular instrument or if this line is significantly absorbed by the surrounding air due to a long optical pass, we recommend using the line at 833.5 nm for carbon determination in low-alloy steels [60].

5. Conclusions

The presented model of stationary plasma source under LTE accounts for the self-absorption, the major line-broadening mechanisms of

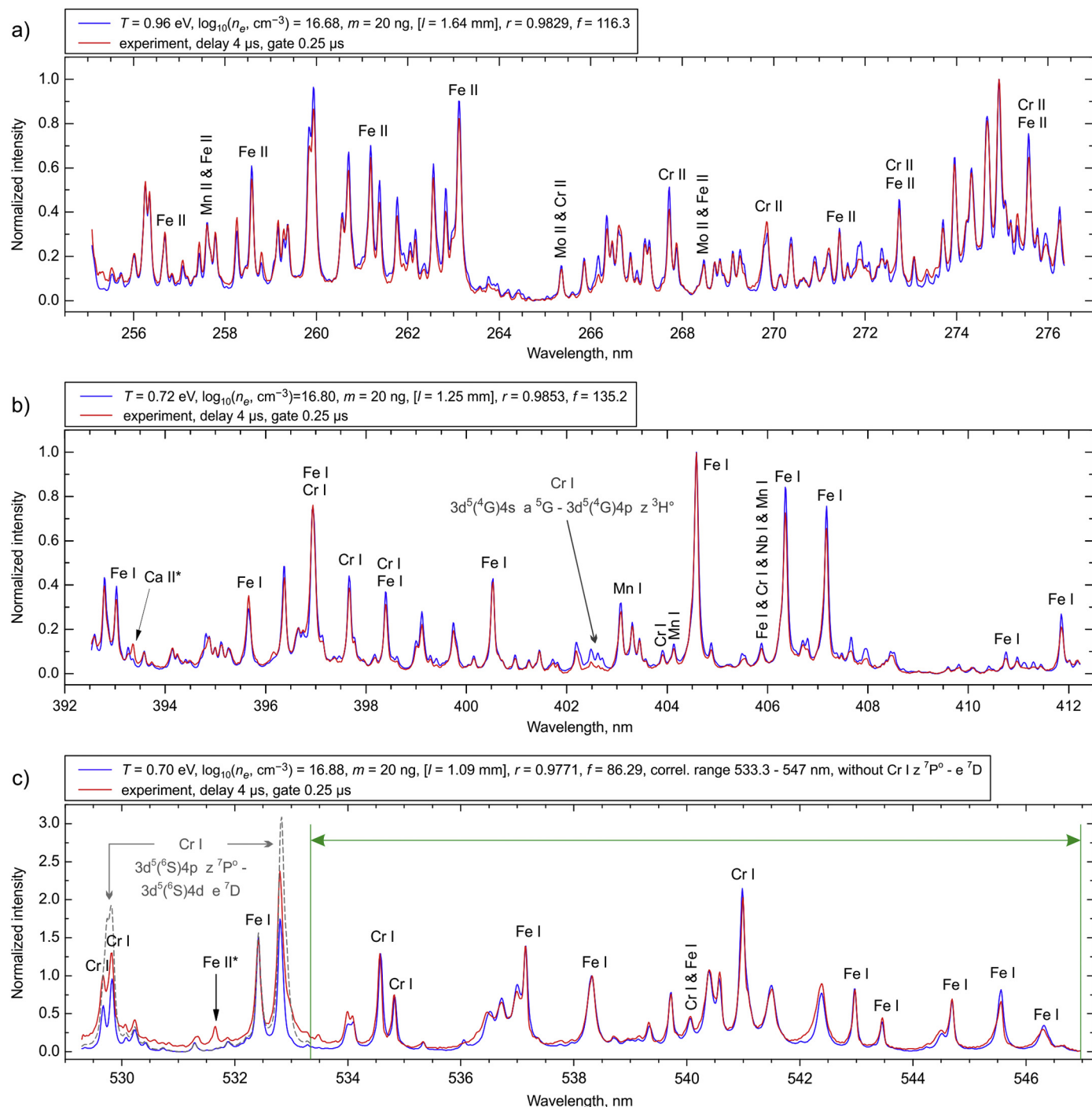


Fig. 8. Experimental (red) and best-fitted model spectra (blue) to ones at the delay 4 μ s for the high-alloy steel within three spectral ranges. The constant background (minimum intensity point) was removed from the spectra; intensities are normalized to selected peaks: (a) at 274.93 nm (Cr II 274.898, Fe II 274.918, Fe II 274.932, Fe II 274.949 lines), (b) 404.58 nm (Fe I 404.581 line), (c) 538.31 (shifted Fe I 538.337 line). The range involved in the fitting procedure in the region 538 nm (plot (c)) is shown by green markers. The shape of the model spectrum within the range 529.3–533.3 nm with the Cr I $z^7P^o - e^7D$ multiplet lines at the plasma parameters given in the legend is shown by the gray dotted line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

lines, and the instrumental distortion of emission spectra. The above factors have a crucial influence on spectra shape. Simultaneously, their calculation leads to significant increase of calculation cost. Nevertheless, through the use of (i) a simple approximation of plasma shape (1D uniform “rod”) providing an analytical solution for the radiative transfer equation, (ii) optimization of the Voigt profile calculation and (iii) the use of FFT for the spectra convolution with the instrumental function we achieved a fairly fast computation of the model

spectra ($\sim 0.4 \mu$ s for calculation of one point in a model spectrum on a conventional PC) allowing spectra fitting even with a regular grid of parameters (T, n_e, m) for a known elemental composition of the plasma. For thorough spectra modeling and line identification, both the NIST and Kurucz spectral data as well as Stark parameters continuously added to our database were used.

A simple cost function based on Pearson's r has a sole maximum in the T – n_e coordinates at a fixed plasma mass. This enabled an easy

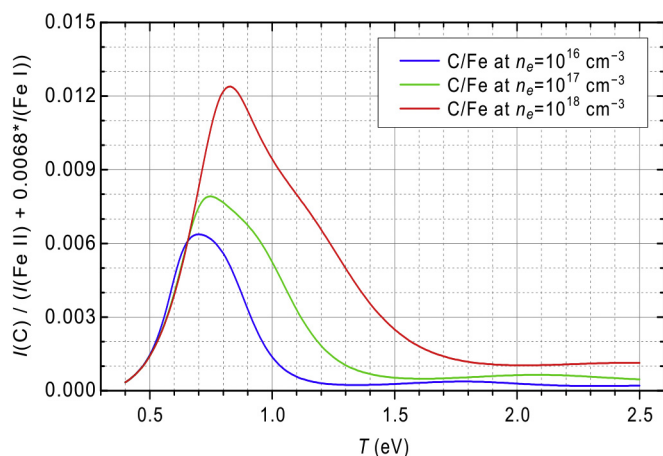


Fig. 9. The calculated intensity ratio of the C I line at 247.856 nm to the interfering iron lines (Fe II 247.857 nm and the wing of the Fe I 247.978 nm at 247.856 nm) at different plasma T and n_e for the low-alloy steel UG112.

finding of the model spectrum that fits the experimental one with the highest similarity. By varying the plasma mass and inherently the degree of self-absorption in the model we can reach, as expected, a strong correlation between the model and experimental spectra when the emission of a particular zone of the plasma is dominated. We found that the best-fitted temperatures differ for the UV and visible spectral ranges due to the different temperatures in the “hot” plasma core and “cold” periphery. The relative differences between the best fitted temperatures of the “hot” and “cold” plasma parts gradually decrease from 40% at a short delay of 1 μ s after the laser pulse to 12% at 10 μ s. The values of electron density evaluated either by modeling or from broadening of the Balmer H_α and Fe I 538.337 nm lines were much closer to each other than the temperatures. The best fitted n_e at medium delays after the laser pulse (4–7 μ s) for the spectra in visible regions was close to the one determined from the Fe I line, while those for the UV region was close to results from H_α line. Generally, a good agreement of our observations with results of Aragón and Aguilera [12,55,56], in which the parameters of plasma resulting from spatially resolved and spatially integrated measurements were compared, proves the adequacy (with certain limitations) of the suggested model and its potential for plasma diagnostics.

We have demonstrated that the accuracy of the experimental data approximation by the model is sufficient for automatic identification of spectra of plasma with known elemental composition. A few discrepancies are primarily related to inhomogeneity of the laser-induced plasma and to the absence of data on Stark broadening parameters. For example, for the high-alloy steel the algorithm identified 106 peaks within the 255–276 nm spectral region. These peaks were associated with the 275 strongest transitions (these small numbers should be compared with the total of 11,700 lines found in the databases). The data on the contribution of a particular line to the observed peak and the evaluation of self-absorption and relative intensity allow preliminary selection of the analytical line. Such data can be preliminary obtained and analyzed at different values of plasma parameters (mainly, temperature) which can be helpful for choosing of a preferred delay for spectra registration. Evaluation of possible spectral interferences from weak lines of major constituents and strong lines of minor and trace elements was very helpful for sensitive determination of light rare earth elements (Y, La) in geosamples, for which the lack of certified reference materials is still a serious problem, with limit of detections below their Earth's crustal abundances [61]. Theoretical modeling supported the assumption that the optimal temporal condition for silver determination in ores (Ag I 328.07 nm line) was achieved at the latest possible delay [43] due to minimal spectral interference from iron lines in the “cold” laser plasma ($T \sim 0.4$ eV). Besides planning of LIBS

analytical studies, the developed tool can be useful for critical evaluation of experimental data. In particular, we have clearly demonstrated that the C I 247.856 nm line cannot provide any analytical information for carbon determination in steels [59]. The modeling made possible re-calibration of a spectrum by its shape only, which resulted in correct lines re-assignment of the lines and in more reliable interpretation of gallstone measurements [62].

In summary, the relatively simple model of laser-induced plasma coupled with spectra-fitting and line-identification algorithms provides fruitful information to plan LIBS studies (either plasma diagnostics or analytical studies) prior to experimental measurement. Simultaneous presence of lines with different upper-level energies, self-absorbed lines, and lines with well-known Stark parameters can provide the most reliable plasma parameters (T , n_e and m). Future development of the model to consider plasma inhomogeneity may improve fitting of spectra containing both atomic and ionic lines and, therefore, improve plasma diagnostics by model spectra.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.sab.2019.06.002>.

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